GLOBAL SCHOOL OF SCIENCE

First Terminal Examination - 2079

Subject: Physics (1021)

Class : XII | **Full Marks: 75**

SET 'A'

Time: 3 hrs. | **Pass Marks: 30**

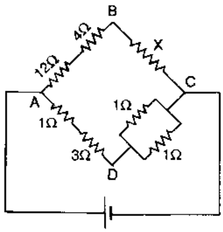
Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group A

Rewrite the best alternative of each question in your answer sheet. [11 × 1 = 11]

1. Wheatstone bridge is used to measure
   a. emf two cells
   b. potential difference
   c. resistance
   d. current
2. The null point in a potentiometer for a cell of emf 1.5 V is obtained at length 300 cm of the potentiometer wire. The potential gradient of the wire is
   a. 0.05 V/m
   b. 0.5 V/m
   c. 0.2 V/m
   d. 2 V/m
3. In the arrangement of resistances shown in fig., the potential difference between B and D will be zero when the unknown resistance X is



   a. 4 Ω
   b. 2 Ω
   c. 3 Ω
   d. emf of cell is needed to find out X.
4. The production of emf by maintaining a difference of temperature between the two junctions of dissimilar metals is called
   a. Joule effect
   b. Seebeck effect
   c. Peltier effect
   d. Thomson effect

5. If only 2% of the main current is to be passed through a galvanometer of resistance G, then the resistance of shunt will be

   a. G/50 b. 50G

   c. 49G d. G/49

6. The internal energy of an ideal gas depends upon

   a. specific volume b. pressure

   c. temperature d. density

7. A heater coil is cut into two equal parts and only one part is used in the heater. How will the heat generated vary?

   a. become one fourth b. halved

   c. doubled d. become four times

8. The time period of a second's pendulum is

   a. 1 sec b. 2 sec

   c. 0.5 sec d. 3 sec

9. An electron and a proton are injected into a uniform electric field at right angles to the direction of the field with equal momentum. Then

   a. the electron trajectory will be less curved than proton trajectory

   b. the proton trajectory will be less curved than electron trajectory

   c. both the trajectories will be equally curved

   d. both the trajectories will be straight

10. Above neutral temperature, thermo e.m.f.

    a. changes sign

    b. increases with the rise in temperature

    c. decreases with the rise in temperature

    d. is constant

11. In rotational motion of a rigid body, all particles move with

    a. same linear and angular velocity

    b. same linear and different angular velocity

    c. different linear velocities and same angular velocities

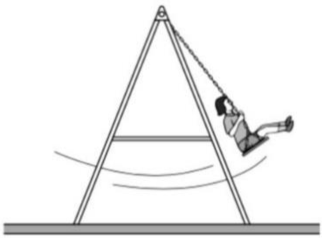
    d. different linear velocities and different angular velocities

Group - B

Give short answer to the following questions. **[8 × 5 = 40]**

12. a. Describe the energy changes that take place as the bob of a simple pendulum makes one complete oscillation, starting at its maximum displacement. [2]

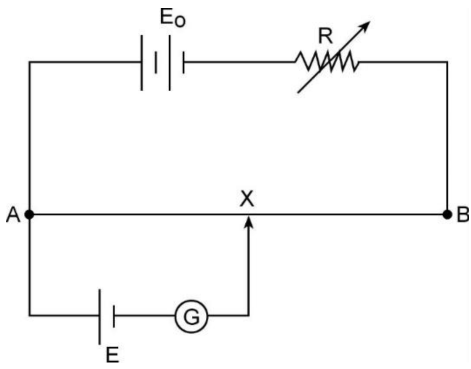
b. Figure shows a young girl swinging on a garden swing. You may assume that the swing behaves as a simple pendulum. Ignore the mass of chains supporting the seat throughout this question, and assume that the effect of air resistance is negligible. 15 complete oscillations of the swing took 42s.



i. Calculate the distance from the top of the chains to the centre of mass of the girl and seat. [2]

ii. Calculate the maximum speed of the girl during the first oscillation if amplitude is 250 mm. [1]

13. A simple potentiometer circuit is shown in figure.



i. State giving reason, how the balance point X is shifted when R is increased? [1]

ii. Why the value of E_o should be greater than E? [1]

iii. The driver cell (E_o) of the potentiometer has an emf of 2V and negligible internal resistance. The potentiometer wire AB has a resistance of 3 Ω. Calculate the resistance R needed in series with the wire if a potential difference of 5.0 mV is required across the whole wire. [3]

14. a. Why a gas has two specific heat capacities? [1]

b. Show that $C_p - C_V = R$. [3]

c. Why change in internal energy is zero in cyclic process. [1]

15. In a Millikan type apparatus, the horizontal plates are 1.5cm apart. With the electric field switched off, the clock oil drop is observed to fall with the steady velocity 25 x 10^{-2} cm/s. When the field is switched on, the upper plate being positive, the drop just remains stationary when the p.d. between the plates is 1500V.

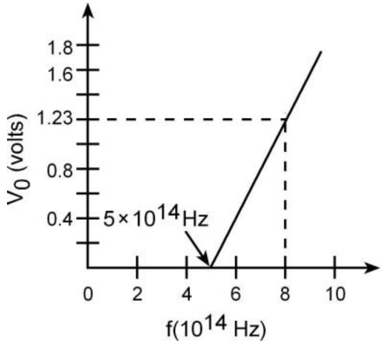
a. Write the condition at which the clock oil drop remains stationary. [1]

b. Calculate the radius of the drop and number of electronic charges. [Density of oil = 900kg/m^3, Viscosity of air = 1.8x10^{-5} kg/ms, neglect density of air] [3]

c. Why clock oil is more preferable? [1]

16. a. R.A. Millikan verified Einstein's photoelectric equation experimentally. Describe an experiment to determine the value of Planck's constant in photoelectric effect. [3]

b. The graph between frequency of incident light (f) and the stopping potential (V_o) is shown in figure.



Use this graph to determine the value of

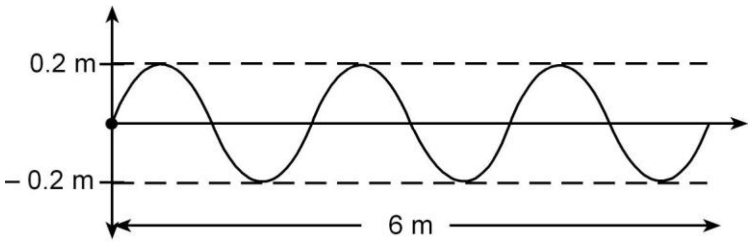
a. threshold frequency

b. Planck's constant [1+1]

17. a. What is the physical meaning of moment of inertia of a rigid body? A ballet dancer stretches her hands when she wants to come to rest. Why? [2]

b. An electric fan is turned off, and its angular velocity decreases uniformly from 500 rev/min to 200 rev/min in 4.00s (i) Find the angular acceleration and the number of revolutions made by the motor in 4.00s interval. (ii) How many more seconds are required for the fan to come to rest if the angular acceleration remains constant? [3]

18. a. The diagram below represents a wave.



i. What is the amplitude and wavelength of the wave? [1]

ii. What is the speed of the wave if the frequency is 8 Hz? [1]

iii. Define phase and write the relation between phase difference and path difference. [1]

b. Write any two differences between longitudinal wave and transverse wave. [2]

19. a. What is Kirchhoff's law? Using Kirchhoff's law, derive the principle of Wheatstone bridge. [3]

b. The thermo-emf 'ε' and the temperature of hot junction 'θ' satisfy a relation $\varepsilon = a\theta + b\theta^2$, where $a = 4.1 \times 10^{-5} V(°C)^{-1}$ and $b = -4.1 \times 10^{-8} V(°C)^{-2}$. If the cold junction temperature is 0°C, find the neutral temperature. [2]

Group - C

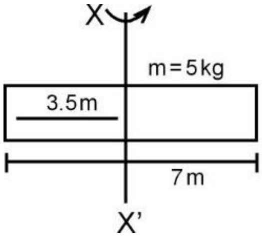
Give long answer to the following questions. **[3 × 8 = 24]**

20. a. A uniform rod of mass 'M' and length '*l*' is rotated about an axis passing through one end of rod and perpendicular to rod. Derive an expression for its moment of inertia and radius of gyration. [3]

b. Explain if the ice on polar caps of the earth melts, how it affects the duration of the day. [1]

c. A constant torque of 200Nm turns a wheel about its centre. The moment of inertia about its axis is $100 kgm^2$. Find kinetic energy gained after 20 revolutions when it starts from rest. [2]

d. A rod of length 7m and mass 5kg is rotated about an axis as shown in figure. Find



i. Moment of inertia of rod

ii. Radius of gyration of rod [2]

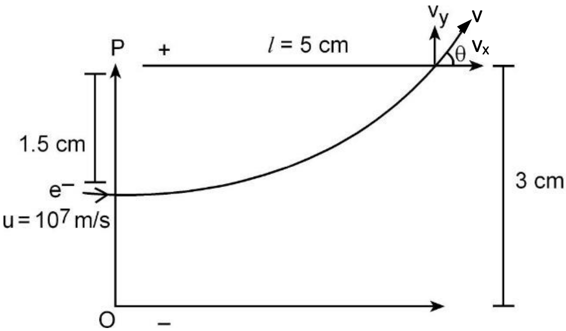
21. a. An electron is projected at right angle to the direction of magnetic field.

i. Show that the nature of path followed by electron is circular. [2]

ii. Obtain an expression of its radius and time period. [2]

b. What will happen if electron is projected at an angle of 45^0 to the direction of magnetic field? Explain. [1]

c. Study the given figure and answer the given questions.



Find: (a) potential difference between plates

(b) value of 'θ'

(c) resultant velocity at the point of emergence. [3]

22. a. What do you mean by a shunt? Describe its use in converting a galvanometer into an ammeter. [1+3]

b. The element of a heater is very hot while the wire carrying current are cold. Why? [1]

c. An electric lamp is rated as 220V- 100W. What is its electrical resistance? What power does it consume if it is used in a circuit of 110V? [3]

- The End -